DA1 - MINI PROJECT QUESTIONS— C Programming (Individual)

Common Rules (Applicable to ALL Projects)
Your project must:
- Use **struct**
- Use **file handling** for permanent storage
- Use **minimum 5 user-defined functions**
- Be **menu-driven**
- Handle **invalid input** (basic validation)
- Must be deployed online (Option A: WASM + GitHub Pages preferred) – Guidelines shared.

Questions

1) Smart Student Performance Analyzer
Problem
Design a system to manage student marks and generate performance insights.
Must Have Features
- Add student record: (RegNo, Name, 3 subject marks)
- Calculate total, average, and grade (A/B/C/Fail)
- Store and load records from file
- Search student by RegNo
- Update marks for a given RegNo
- Display topper(s)
- Class statistics: class average, highest total, lowest total
- Generate **grade distribution** (how many A/B/C/Fail).

2) Mini Banking System with Transaction Log
Problem
Create a simple banking console system with account operations and transaction history.
Must Have Features
- Create account: (AccNo, Name, Balance)
- Deposit & withdraw (prevent negative balance)
- Save accounts in file
- Maintain a **transaction log file** (AccNo, type, amount, timestamp/date)
- View account summary
- Search account by AccNo
- Display the **last 5 transactions** for an account.

3) Inventory Management with Low-Stock Alert
Problem
Build an inventory system used by a small shop.
Must Have Features
- Add product: (ProductID, Name, Price, Quantity)
- Update stock (purchase/sale)
- Delete a product
- Search by ProductID
- Save/load from file
- Low-stock report: list items below threshold
- Show total inventory value = Σ(price × quantity).

4) Library Book Tracking System
Problem
Track books, issue/return, and availability.
Must Have Features
- Add book: (BookID, Title, Author, Status)
- Issue book to student (store issued-to RegNo)
- Return book
- Search book by ID or title keyword
- Save/load records from file
- Prevent issuing if already issued; show list of currently issued books.

5) Expense & Budget Tracker (Personal Finance)
Problem
Help users track expenses and compare with monthly budget.
Must Have Features
- Add expense: (Date, Category, Amount, Note)
- Store in file and load
- Show total expense for selected month
- Category-wise total (Food/Travel/Medical/etc.)
- Search expenses by date or category
- Budget limit: warn if monthly total exceeds budget.

6) Attendance Manager with Shortage Alert
Problem
Track attendance and identify shortage students.
Must Have Features
- Add student: (RegNo, Name)
- Mark attendance for a given date (P/A)
- Store attendance in file
- Compute attendance % per student
- List students below 75%
- Generate a summary report: class average attendance.

7) Employee Payroll System
Problem
Compute salaries and generate payroll report.
Must Have Features
- Add employee: (EmpID, Name, BasicPay, OT Hours)
- Gross pay = Basic + (OT Hours × OT Rate)
- Tax deduction slab (simple rules)
- Net pay calculation
- Save/load records
- Generate payslip-style output for a given EmpID.

8) Hospital Patient Registration & Follow-up Tracker
Problem
Track patient visits and diagnosis notes for a small clinic.
Must Have Features
- Register patient: (PatientID, Name, Age, Phone)
- Add visit record: (PatientID, Date, Diagnosis, Prescription note)
- Save/load patient + visits in file
- Search patient by ID
- Show visit history for patient
- Show count of visits per patient and identify “frequent visitors”.

9) Movie Ticket Booking System (Seat-based)
Problem
Build a simple movie ticket booking system for one theatre.
Must Have Features
- Display available movies/shows (hardcode 3–5 shows OR load from file)
- Seat layout (example: 5 rows × 10 seats = 50 seats)
- Book tickets: choose show + number of seats + seat numbers
- Prevent double booking (seat once booked cannot be booked again)

- Generate booking receipt: BookingID, customer name, show, seats, total amount
- Store bookings in file
- View booking by BookingID
- Show “occupancy report” per show:
  - total seats
  - booked seats
  - available seats
  - occupancy %

10) Bus Ticket Reservation + Cancellation System
Problem
Create a bus reservation system with booking and cancellation.
Must Have Features
- Create bus routes (2–3 routes) with fare and seat capacity
- Book seat(s): passenger name, route, seat number(s)
- Generate TicketID and store booking in file
- Display available seats for a route
- Cancel ticket by TicketID (free the seat)
- Search ticket details by TicketID
- Daily collection report:
  - total tickets booked per route
  - total revenue per route

Deliverables (What students must submit)
1. **Live hosted link** (GitHub Pages WASM preferred)
2. **PDF documentation** – Template shared
- Cover page with links
- Problem statement + features
- Struct + function list table
- File format description
- Testing table (min 6 test cases)
- Screenshots (min 3)
- AI usage declaration